Shaft Component Generator (Version: 2025.2 (Build 292293000, 293))

5/28/2025

Project Info (iProperties)

Calculation

Material

Material		AISI 1050
Modulus of Elasticity	E	210000 MPa
Modulus of Rigidity	G	81000 MPa
Density	ρ	7850 kg/m^3

Calculation Properties

Include			
Yes	Density	ρ	7850 kg/m^3
Yes	Shear Displacement Ratio	β	1.188 ul
	Number of Divisions		1000 ul
	Mode of reduced stress		HMH

Loads

Index	Location	Radial Force				Bending Moment				Continuous Load					Axial Force	Torque	Deflection				Deflection Angle
		Y	X	Size	Direction	Y	X	Size	Direction	Y	X	Size	Direction	Length			Y	X	Size	Direction	
1	190 mm	-3726.340 N	10238.000 N	10895.056 N	110.00 deg												3.528 microm	-9.908 microm	10.518 microm	289.60 deg	0.00 deg

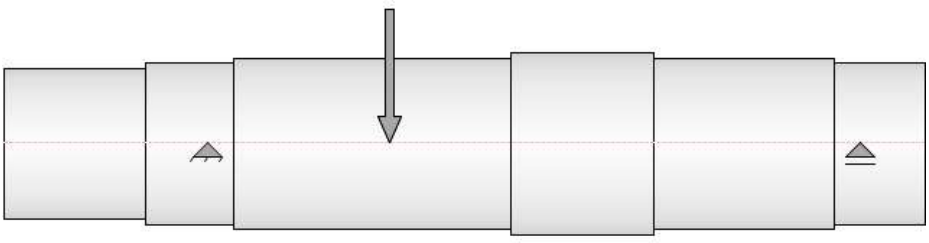
Supports

Index	Type	Location	Reaction Force					Yielding	Type	Deflection				Deflection Angle
			Y	X	Size	Direction	Axial Force			Y	X	Size	Direction	
1	Fixed	100 mm	-2573.412 N	7376.447 N	7812.453 N	109.23 deg			User	-0.000 microm	-0.000 microm	0.000 microm	180.00 deg	0.01 deg
2	Free	422 mm	-962.063 N	2861.553 N	3018.948 N	108.58 deg			User	-0.000 microm	-0.000 microm	0.000 microm	184.23 deg	0.01 deg

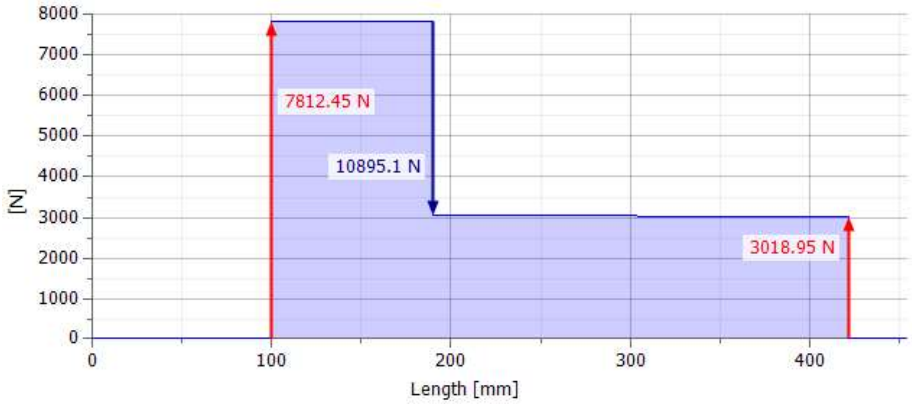
Results

Length	L	454.000 mm
Mass	Mass	19.463 kg
Maximal Bending Stress	σ_B	11.699 MPa
Maximal Shear Stress	τ_S	1.557 MPa
Maximal Torsional Stress	τ	0.000 MPa
Maximal Tension Stress	σ_T	0.000 MPa
Maximal Reduced Stress	σ_{red}	11.941 MPa
Maximal Deflection	f_{max}	11.629 microm
Angle of Twist	φ	0.00 deg

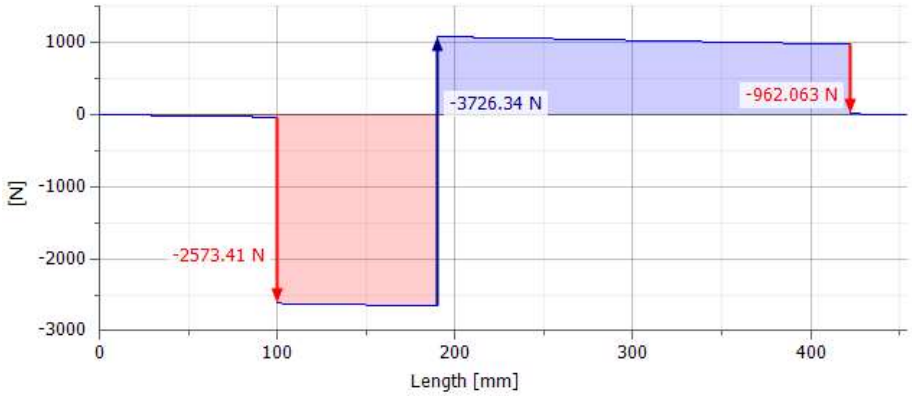
Preview



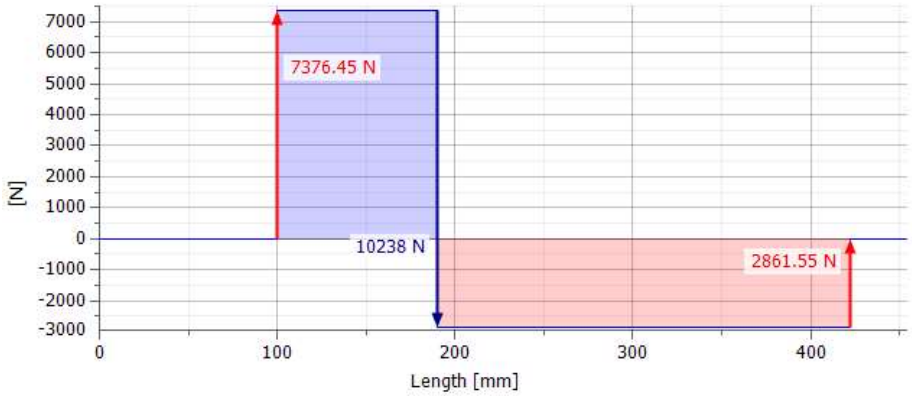
Shear Force



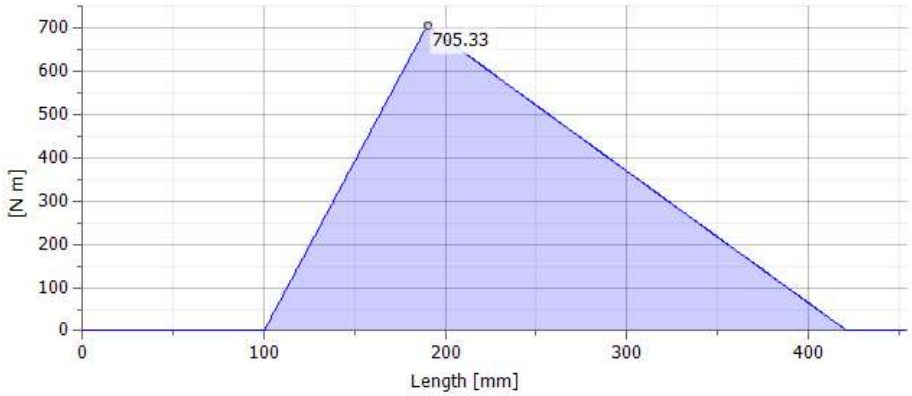
Shear Force, YZ Plane



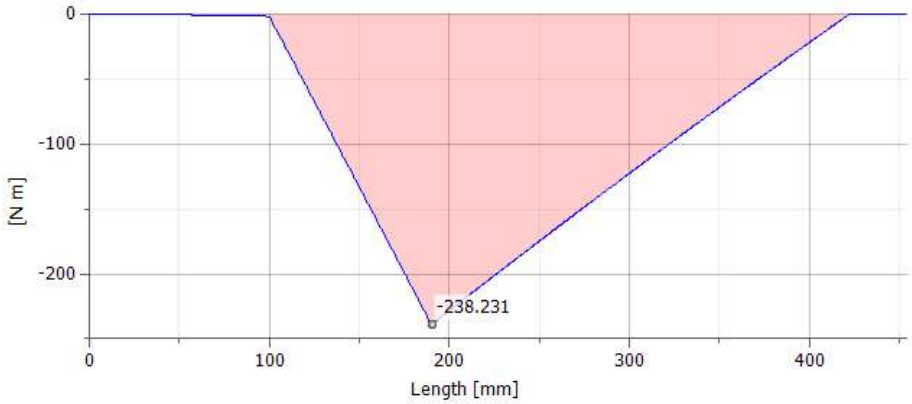
Shear Force, XZ Plane



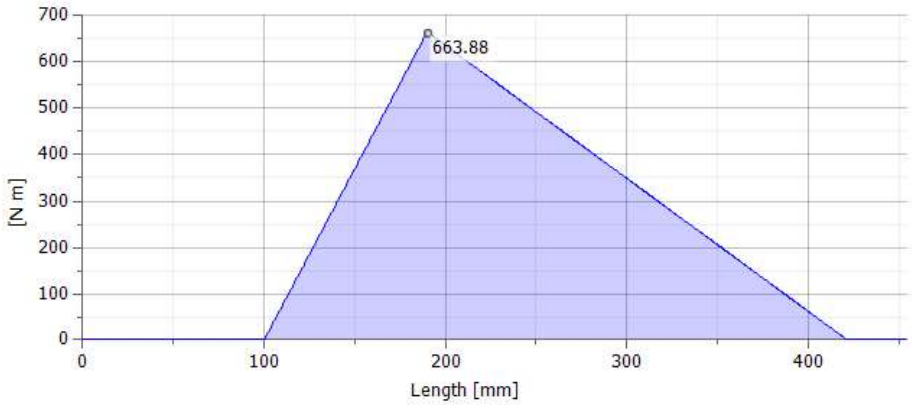
Bending Moment



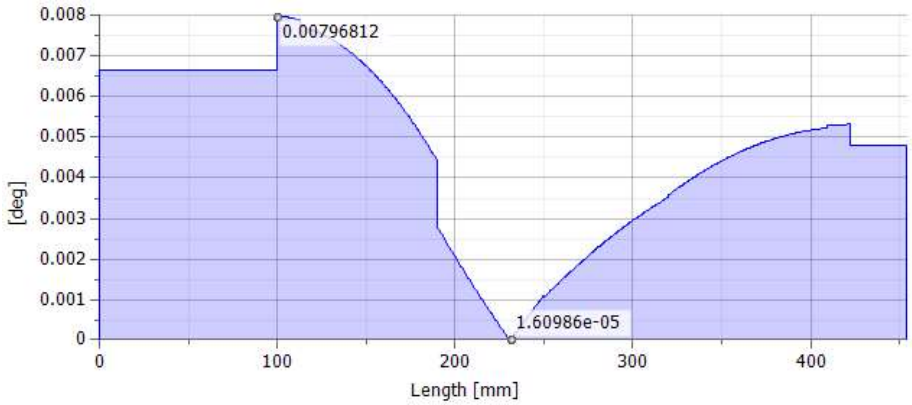
Bending Moment, YZ Plane



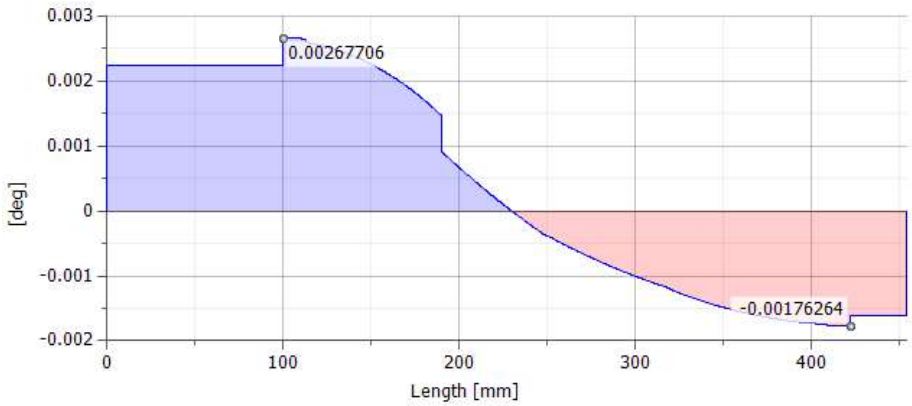
Bending Moment, XZ Plane



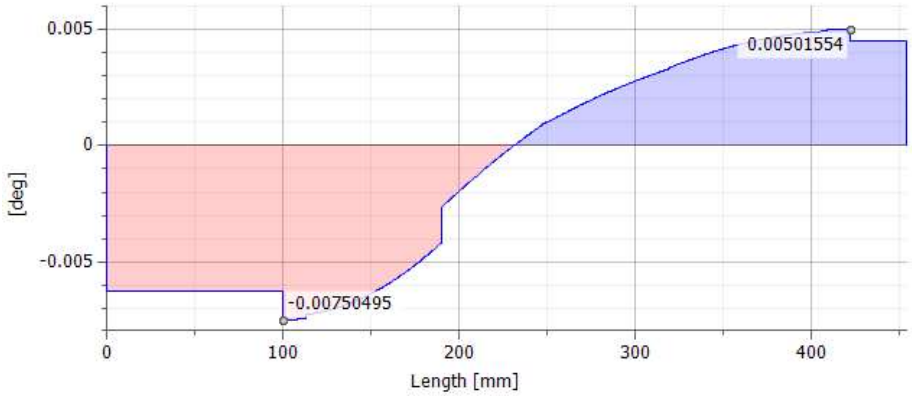
Deflection Angle



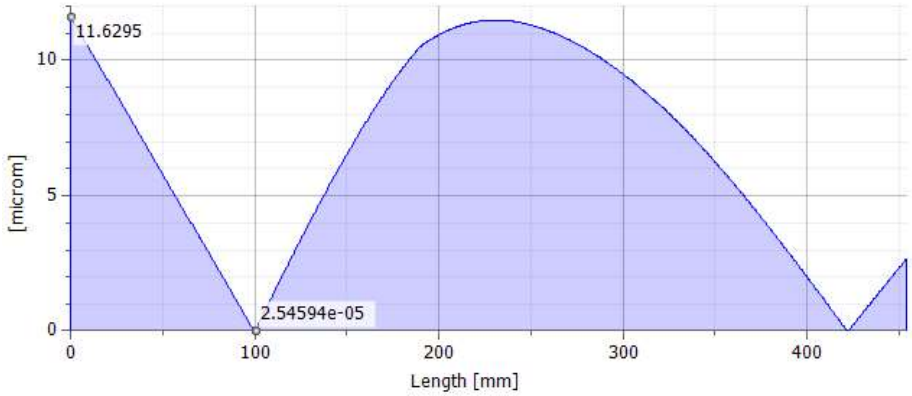
Deflection Angle, YZ Plane



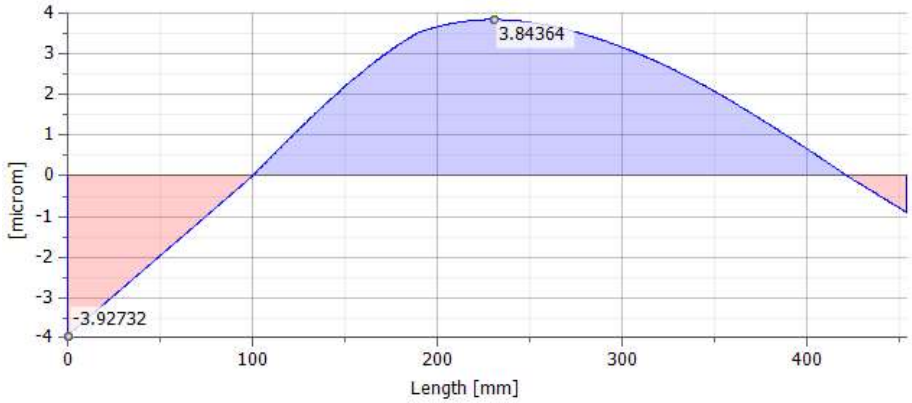
Deflection Angle, XZ Plane



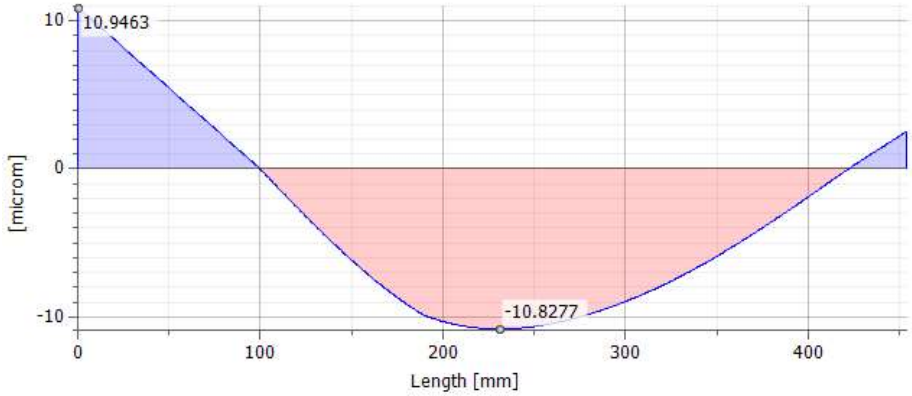
Deflection



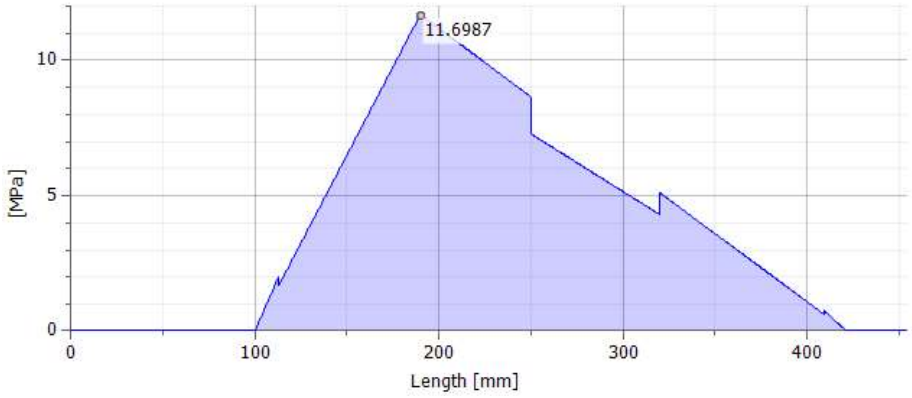
Deflection, YZ Plane



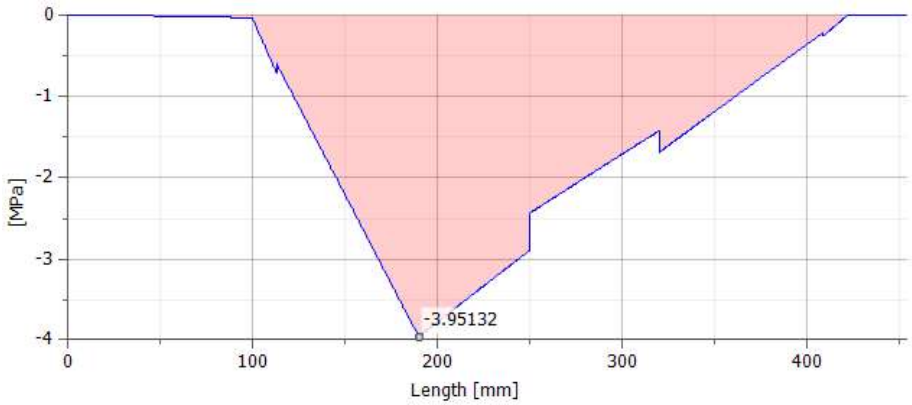
Deflection, XZ Plane



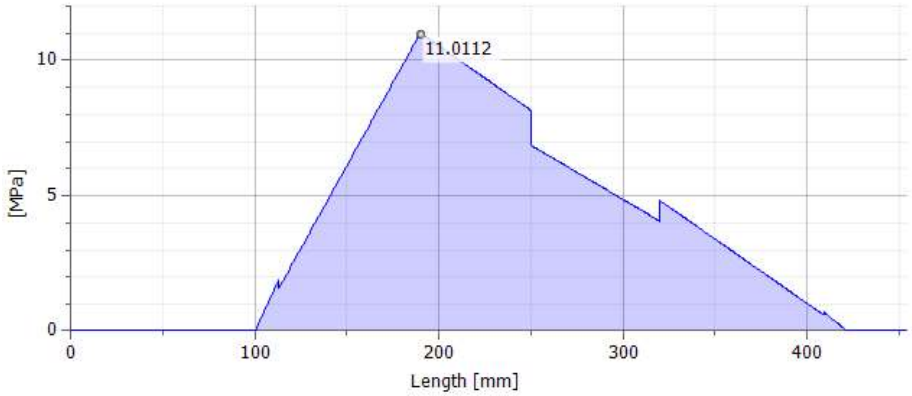
Bending Stress



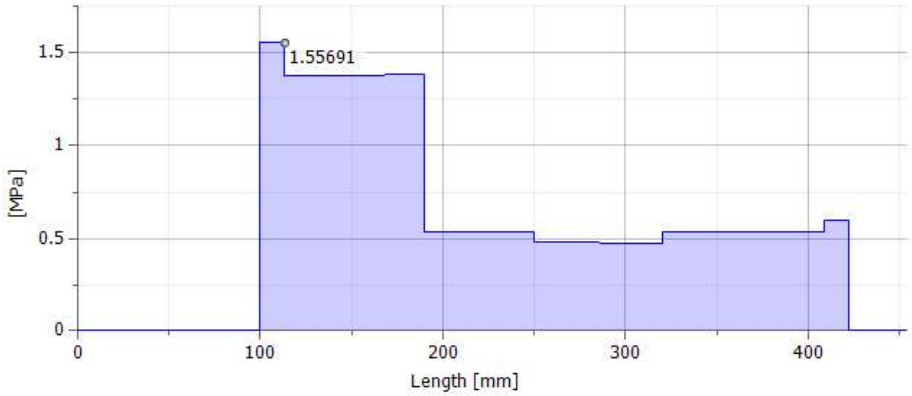
Bending Stress, YZ Plane



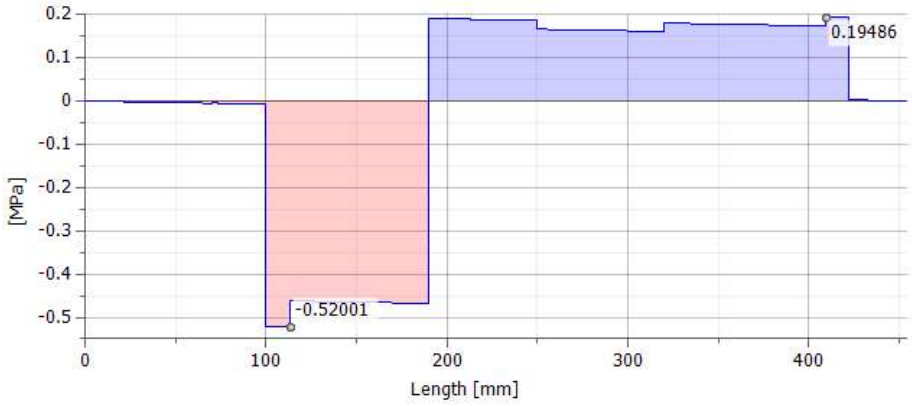
Bending Stress, XZ Plane



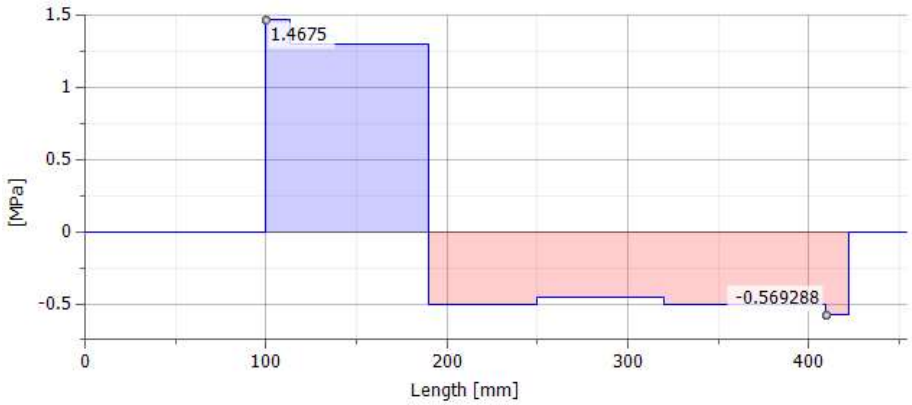
Shear Stress



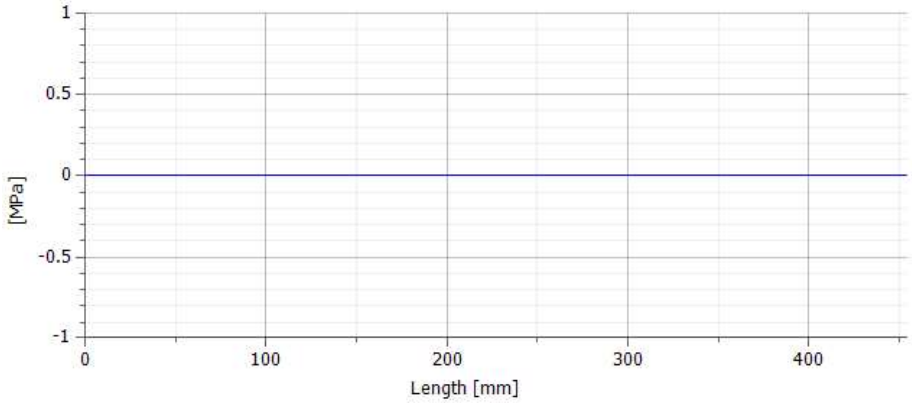
Shear Stress, YZ Plane



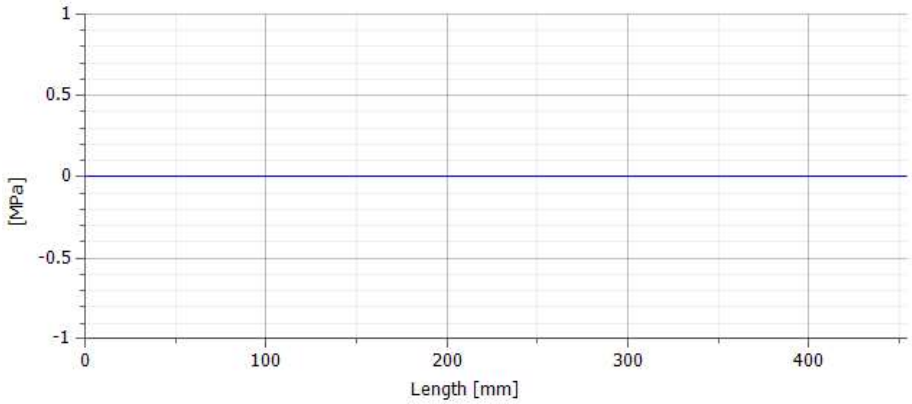
Shear Stress, XZ Plane



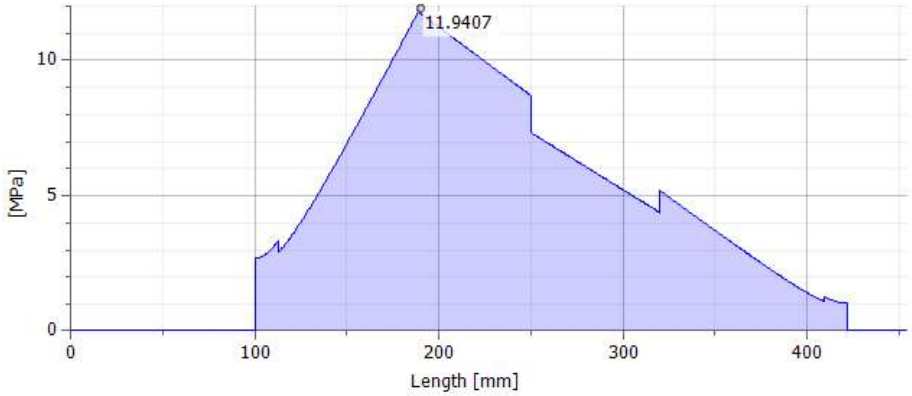
Torsional Stress



Tension Stress



Reduced Stress



Ideal Diameter



Summary of Messages

7:10:14 AM Calculation: Calculated.